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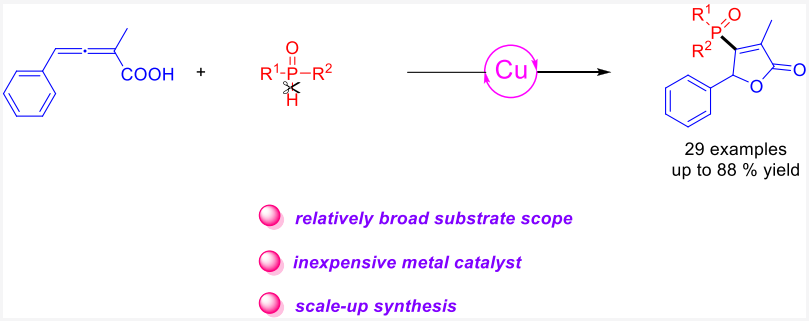
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Copper-Catalyzed Phosphorylation of 2,3-Allenoic Acids and Phosphine Oxide: Access to Phosphorylated Butenolides

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ABSTRACT: We investigated a novel Cu-catalyzed annulation of 2,3-allenoic acids with diphenylphosphine oxide, leading to the formation of 4-phosphate butenolides in up to 88% yield. The formation of the C−P bond provides new avenues for the functionalization of different furan-2(5*H*)-ones, with favorable features such as suitable functional group tolerance and mild synthesis conditions.

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INTRODUCTION

As effective scaffolds, furan-2(5*H*)-ones are widespread in natural products, bioactive molecules, and drugs (Figure 1).[1] Therefore, developing a valid and efficient method for their synthesis represents a key goal in organic synthesis.[2] For instance, Ma et al. originally developed a method for the cyclization of halogenated metals and allenoic acids to afford *β*-halobutenolides (Scheme 1a).[3] Subsequent studies have achieved C−C bond formation of allenoic acids with aryl iodides,[4] alkenes,[5] alkynes,[6] or allenes[7] to form polysubstituted butenolides (Scheme 1b). Afterward, various C−S bond formation reactions, including trifluoromethylthiolation, sulfenylation, sulfonylation, and selenylation have been widely reported (Scheme 1c).[8] Despite these remarkable advances, the exploration of other functional groups to prepare different types of functionalized butenolides remains a critical task.

Phosphorus-containing heterocycles are an important class of organophosphorus compounds, applied widely in synthetic[9] and medicinal chemistry,[10] as well as material science[11] and universal ligands,[12] drawing considerable attention for both industrial and medicinal applications. Generally, the traditional synthesis of phosphorus-substituted heterocycles is based on direct phosphorylation of the parent heterocycles.[13] Various effective C−P bond reactions involving the addition of P-centered radicals to initiate the formation of the heterocycle have recently been developed.[14] It is worth noting that this kind of reaction is efficient and functional as the phosphorus-substituted heterocycle is formed with the concomitant combination of a phosphorus portion. However, only few methods for the introduction of phosphorus substituents into furan-2(5*H*)-ones have been reported. Therefore, the identification of a method for preparing phosphorus-containing furanones is a highly valuable task. Based on our previous investigation of radical synthesis,[15] in this study, we report the preparation of a phosphorus-containing furanone via the highly efficient Cu-catalyzed annulation of allenoic acids with phosphorus, which directly forms C−P bonds and creates furanone rings in one reaction. (Scheme 1d).

RESULTS AND DISCUSSION

As shown in Table 1, the radical phosphorylation of 2-methyl-4-phenyl-2,3-allenoic acid (**1a**) was chosen as a model reaction, with diphenylphosphine oxide (**2a**) as the phosphorus source. The reaction was carried out in the presence of $FeCl_2$ (20 mol %) with $K_2S_2O_8$ (2 equiv) as the oxidant in the CH_3CN/HOAc solution, providing the desired product in 17% yield. Among the various screened catalysts (entries **1**−**7**), $Cu(OAc)_2$ was proved to be the most suitable one, giving the product in 80% yield (entry **5**). Other oxidants such as TBHP,

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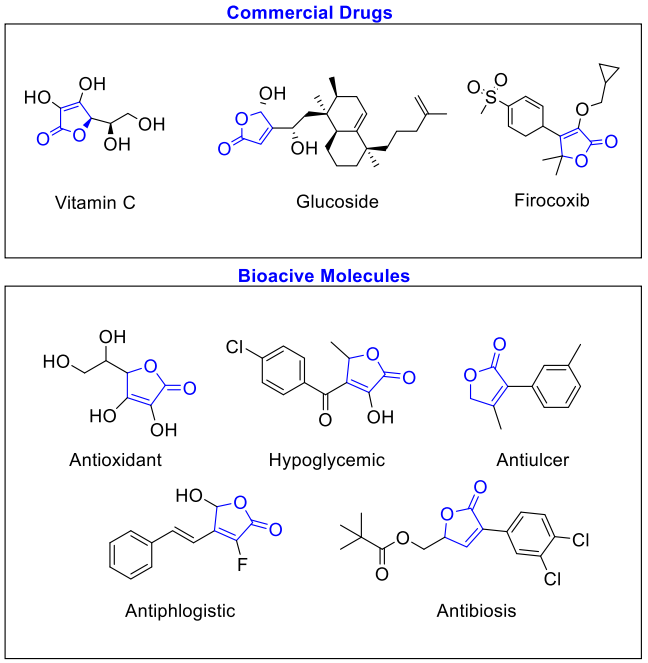


Figure 1. Commercial drugs and bioactive molecules containing furan-2(5*H*)-ones.

$(NH_4)_2S_2O_8$, and $Na_2S_2O_8$ led to lower yields (20−35%) (entries **8**−**10**). Based on these results, it was critical for the success of the reaction to use an acid as the cosolvent (entry **11**). On the other hand, using formic acid instead of HOAc or adjusting the CH_3CN/HOAc ratio did not improve the efficiency of the reaction (entries **12**−**14**). Lastly, further decrease in **2a** concentration and reaction temperature resulted in lower yields (entries **15**, **16**). Therefore, the optimal reaction conditions were as follows: **1a** (0.2 mmol), **2a** (0.4 mmol), $K_2S_2O_8$ (0.4 mmol), $Cu(OAc)_2$ (20 mol %), and CH_3CN/HOAc = 3:1 (2 mL).

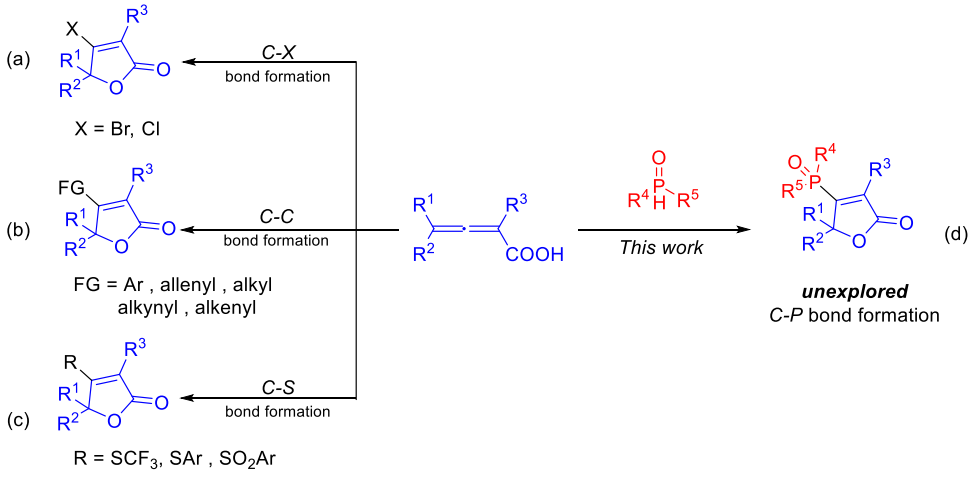
We investigated the range of allenes substrates after determining the optimal reaction conditions (Scheme 2). Benzene rings bearing electron-donating groups, such as -CH_3, *t*-$C(CH_3)_3$, and -OCH_3, at the *para* position reacted with the diphenylphosphine oxide **2a** to deliver the phosphated butenolides **3b**−**3d** in 42−70% yields. On the other hand, functional groups such as fluoro, chloro, bromo, and trifluoromethyl in the *para* position of the aryl ring were well tolerated, and the products **3e**−**3h** were acquired in 48−82% yields. Substrates present in the meta (**3i**) and *ortho* (**3j**−**3l**) positions of the aryl groups reacted smoothly, providing the corresponding products in 37−77% yields. The desired products **3m** and **3n** were provided in high yields both electron-deficient (CF_3) groups and electron-donating (Me) on the phenyl ring. In addition, the substrate with naphthyl as the R^1 substituent could also react with diphenylphosphine oxide (**2a**) to afford the corresponding product **3o** in moderate yield (51%); the slight decrease in yield might be due to steric hindrance effects. It is rather remarkable that this type of reaction could also be utilized for the substrate with *n*-propyl as the R^3 substituent (**3p**). Similarly, both Ph and Me groups at the R^2 position led to lower yields (**3q** and **3r**) because of steric hindrance.[14i]

To further expand the versatility of the phosphorylated reaction, we next explored the range of **2** using the optimized reaction conditions shown in Scheme 3. First, different aromatic substituents (−CH_3, −OMe, −F, −Cl, −Br, and −Ph) were found to be compatible with the optimized conditions, giving the final products **4a**−**4g** in moderate to excellent yields (58−88%). However, the disubstituted substrate **4h** was found to be less tolerated in the present reaction, giving 43% yield. The reaction was further extended to other *H*-phosphonate, disopropyl, and diethyl phosphites that reacted well with allenoic acids (**1a**), generating the corresponding products **4i**−**4j** in 31−33% yields. Interestingly, the reaction of the asymmetric *H*-phosphinate ester proceeded smoothly, providing product **4k** in 41% yield.

In order to investigate the potential applications of the present approach, a gram-scale experiment was carried out, providing the corresponding product **3a** in 62% yield (Scheme 4). Moreover, the ring-opening reaction of **3a** with *t*-BuOH delivered the 1,4-dicarbonyl compound in 42% yield, which is an important precursor and intermediate in organic synthesis.[16]

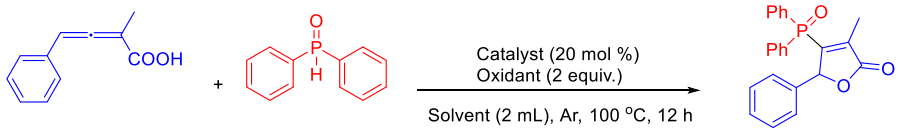
A series of control experiments were conducted to better understand the phosphorylation mechanism (Scheme 5). Although the reaction was completely suppressed when the radical trapping reagents (2,2,6,6-tetramethylpiperidin-1-yl)-oxyl (TEMPO) and butylated hydroxytoluene (BHT) were

Scheme 1. Synthesis of Various Butenolides



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Table 1. Optimization of Reaction Conditions[a]



1a 2a 3a

entry	catalyst	oxidant	solvent	yield[b] (%)
1	$FeCl_2$	$K_2S_2O_8$	CH_3CN:HOAc = 3:1	17
2	CuI	$K_2S_2O_8$	CH_3CN:HOAc = 3:1	n. r.
3	$Fe(OAc)_2$	$K_2S_2O_8$	CH_3CN:HOAc = 3:1	27
4	CuCl	$K_2S_2O_8$	CH_3CN:HOAc = 3:1	trace
5	***$Cu(OAc)_2$***	***$K_2S_2O_8$***	***CH_3CN:HOAc = 3:1***	***80***
6	$CuBr_2$	$K_2S_2O_8$	CH_3CN:HOAc = 3:1	37
7	$Cu(CH_3CN)_4BF_4$	$K_2S_2O_8$	CH_3CN:HOAc = 3:1	66
8	$Cu(OAc)_2$	$(NH_4)_2S_2O_8$	CH_3CN:HOAc = 3:1	57
9	$Cu(OAc)_2$	$Na_2S_2O_8$	CH_3CN:HOAc = 3:1	45
10	$Cu(OAc)_2$	TBHP	CH_3CN:HOAc = 3:1	59
11	$Cu(OAc)_2$	$K_2S_2O_8$	CH_3CN	47
12	$Cu(OAc)_2$	$K_2S_2O_8$	CH_3CN:HCOOH = 3:1	68
13	$Cu(OAc)_2$	$K_2S_2O_8$	CH_3CN:HOAc = 1:1	73
14	$Cu(OAc)_2$	$K_2S_2O_8$	CH_3CN:HOAc = 5:1	69
15[c]	$Cu(OAc)_2$	$K_2S_2O_8$	CH_3CN:HOAc = 3:1	69
16[d]	$Cu(OAc)_2$	$K_2S_2O_8$	CH_3CN:HOAc = 3:1	65

[a]Reaction conditions: 1a (0.2 mmol), 2a (0.4 mmol), catalyst (20 mol %), oxidant (2 equiv), and solvent (2.0 mL) at 100 °C for 12 h under Ar. [b]Isolated yield. [c]The reaction temperature was 80 °C. [d]$Ph_2P(O)H$ (1.1 equiv).

added to the reaction system under standard reaction conditions, a BHT-**2a** adduct was detected by high-resolution mass spectrometry (HRMS), suggesting that a radical pathway might be involved in the process (Scheme 5a,b). In the absence of $K_2S_2O_8$, no phosphorylated product was delivered, which proved the important role of the oxidant in radical cyclization reaction (Scheme 5c). In addition, this reaction could occur without the copper catalyst, giving product **3a** in a limited yield (Scheme 5d). On the other hand, the reaction of $AlCl_3$ and $ZnCl_2$, instead of $Cu(OAc)_2$, as Lewis acids did not yield the desired product **3a**. These results indicated that $Cu(OAc)_2$ may not play the role of a Lewis acid (Scheme 5e). Finally, when **6** reacted with the H−P(O) compound **2a** under standard conditions, product **3a** was not formed. These results clearly indicate that 2,3-allenoic acids may not first undergo intramolecular cyclization to form **6** and then react with the phosphoryl radical (Scheme 5f).

Based on the control experiments discussed above and previous studies, we believe that the present reaction may proceed through the free radical pathway illustrated in Scheme 6. First, in the presence of $K_2S_2O_8$, diphenylphosphine oxide undergoes a deprotonation process to generate a P-centered radical (Path A). In path B, Cu(II) oxidizes the tautomer of diphenylphosphine oxide and deprotonates to generate the radical, while $K_2S_2O_8$ oxidizes Cu(I) to obtain Cu(II).[17] Subsequently, the phosphoryl radical attacks 2,3-allylic acid (**1a**) to obtain the free radical intermediate **A**. Next, the oxidation of the free radical species generates the allyl cation **B**. Lastly, the intermediate **B** provides the phosphorylated product **3** through the process of intramolecular nucleophilic attack.

CONCLUSIONS

In summary, a Cu-catalyzed phosphorylation of 2,3-allenoic acids has been explored, providing a series of 4-phosphate butenolides in up to 88% yield. The meaningful features of this approach include mild reaction conditions, inexpensive metal catalysts, broad substrate applicability, and high functional group tolerance. Importantly, all the obtained phosphated butenolides have not been previously synthesized and might find potential applications in medicinal chemistry. Therefore, this method provides a superior alternative to the formation of C sp^2−p bonds for the direct synthesis of furan-2(5*H*)-ones.

EXPERIMENTAL SECTION

General Methods. Unless otherwise noted, all commercially available sources were used as received. The visualization was carried out under ultraviolet (UV) light (254 nm). The Büchi B-540 capillary melting point apparatus was used for the determination of the melting point, and NMR spectra were recorded on a Bruker Ascend 400 MHz spectrometer operating at 100 MHz of ^{13}C, 400 MHz of ^{1}H, 162 MHz of ^{31}P, and 376 MHz of ^{19}F. Chemical shifts (δ) are measured downfield from tetramethylsilane (TMS) (= 0.0) for ^{1}H NMR, or from $CDCl_3$ (= 77.0) for ^{13}C NMR. HRMS spectra were recorded on an ESI quadrupole time-of-flight mass spectrometer.

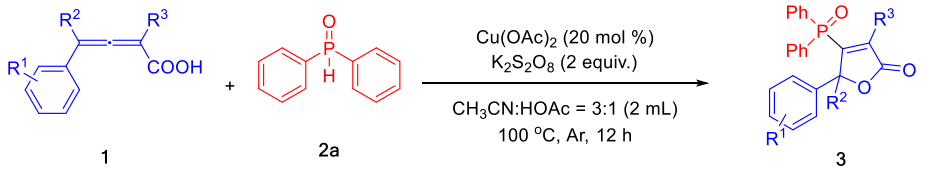
General Procedure for the Synthesis of Compounds furan-2(5*H*)-ones. Synthesis of 3a is representative. 2,3-Allenoic Acids **1a** (34.8 mg, 0.2 mmol), **2a** (0.4 mmol, 2 equiv), $Cu(OAc)_2$ (0.04 mmol, 20 mol %), $K_2S_2O_8$ (0.4 mmol, 2 equiv), CH_3CN (1.5 mL), and HOAc (0.5 mL) were mixed. The mixture was placed in a preheated metal block and stirred at 100 °C (temperature of the metal block) under an argon atmosphere for 12 h. Then, the resulting mixture was cooled to room temperature and concentrated under reduced pressure. The residue was purified by flash chromatography (petroleum ether/ethyl acetate = 2:1) to give **3a** (59.9 mg, 80% yield) as a white solid.

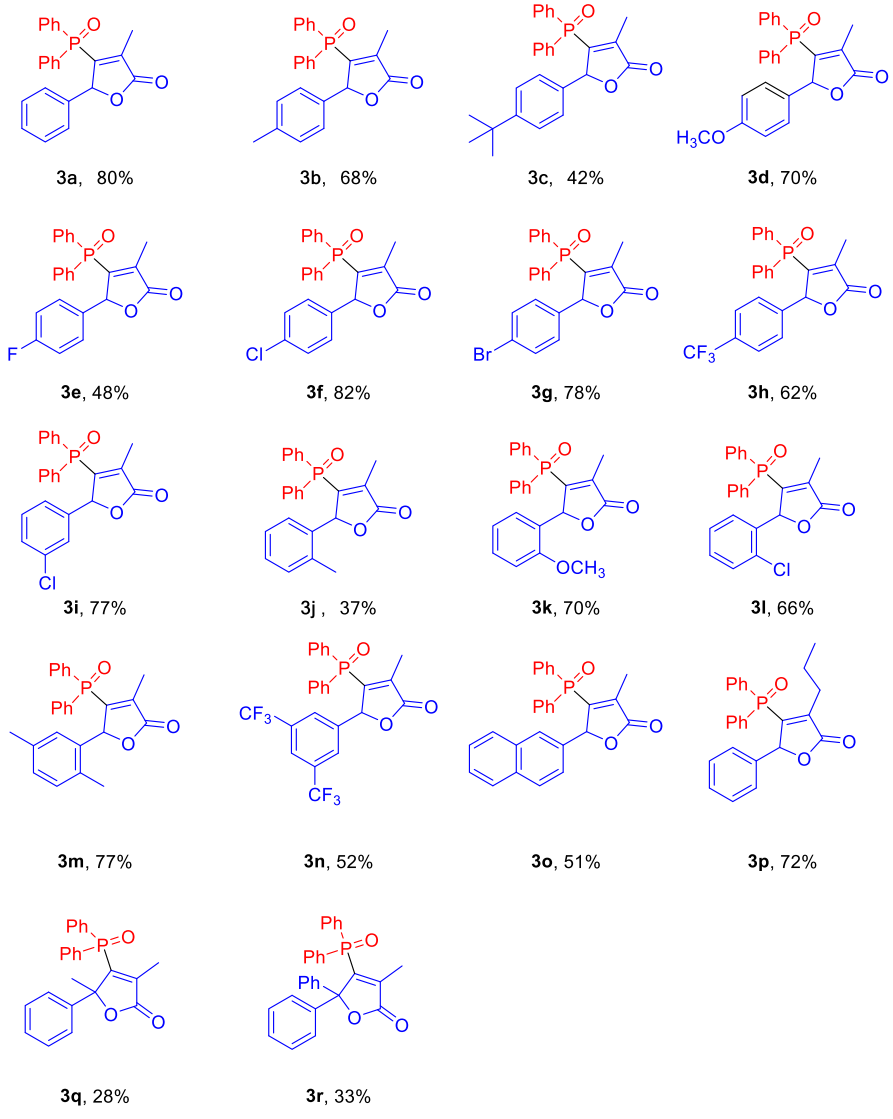
Preparation of Starting Materials. (**1a-1g, 1i-1l, 1o**),[7c,18b] (**1h, 1m, 1n**),[8a] **1p-1r**,[18a] **2b**,[19b] **2c-2h**[19a] were prepared using the literature procedures. In addition, **2a, 2i-2l** were commercially available.

Gram-Scale Synthesis of 3a. To a three-neck flask were added **1a** (1.134 g, 6 mmol), **2a** (2.426 g, 12 mmol), $Cu(OAc)_2$ (0.218 g, 20 mol %), $K_2S_2O_8$ (3.243 g, 2 equiv), CH_3CN (45.0 mL), and HOAC

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Scheme 2. Substrate Scope Relative to 2,3-Allenoic Acids[ab]





[a]Reaction conditions: **1** (0.2 mmol), **2a** (0.4 mmol), $Cu(OAc)_2$ (20 mol %), $K_2S_2O_8$ (2 equiv.) and CH_3CN: HOAc = 3:1 (2.0 mL) at 100 °C for 12 h under Ar. [b]Isolated yield.

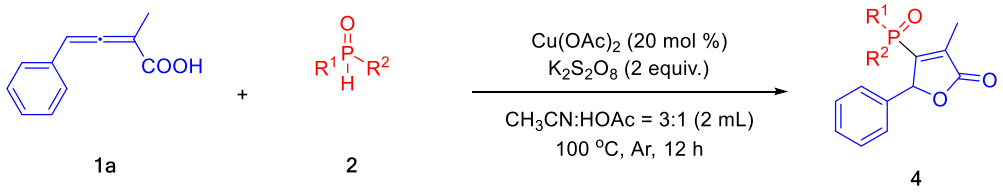
(15.0 mL). After stirring at 100 °C (temperature of the oil bath) under Ar for 12 h, the mixture was concentrated under reduced pressure. The residue **3a** as a white solid (1.392 g, 62% yield) was purified by flash chromatography (PE/EA = 2:1).

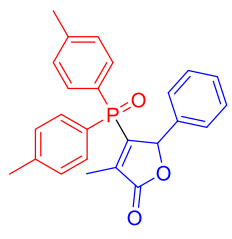
*4-(diphenylphosphoryl)-3-methyl-5-phenylfuran-2(5H)-one (**3a**).* The product **3a** was obtained as a white solid (59.9 mg, 80% yield, m.p. 168−169 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.57−7.34 (m, 10H), 7.17 (t, J = 7.2 Hz, 1H), 7.09 (t, J = 7.6 Hz, 2H), 6.87 (d, J = 7.0 Hz, 2H), 6.02 (s, 1H), 1.88 (t, J = 2.0 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 172.8 (d, J = 20.5 Hz), 153.1 (d, J = 97.2 Hz), 139.6 (d, J = 7.7 Hz), 132.7, 132.6 (d, J = 2.8 Hz), 132.5 (d, J = 2.8 Hz), 131.2 (d, J = 10.6 Hz), 131.0 (d, J = 10.3 Hz), 130.1 (d, J = 108.6 Hz), 129.4 (d, J = 108.6 Hz), 129.1, 128.8 (d, J = 4.8 Hz), 128.7 (d, J = 4.8 Hz), 128.4, 127.4, 84.9 (d, J = 14.1 Hz), 11.3 (d, J = 2.3 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 19.22. HRMS (EI) *m/z*: calcd for $C_{23}H_{20}O_3P$ $[M]^+$, 375.1145; found, 375.1152.

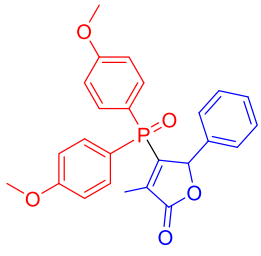
*4-(diphenylphosphoryl)-3-methyl-5-(p-tolyl)furan-2(5H)-one (**3b**).* The product **3b** was obtained as a white solid (52.8 mg, 68% yield, m.p. 173−174 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.56−7.33 (m, 10H), 6.88 (d, J = 7.8 Hz, 2H), 6.73 (d, J = 8.0 Hz, 2H), 5.97 (s, 1H), 2.23 (s, 3H), 1.87 (t, J = 2.2 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 172.8 (d, J = 20.6 Hz), 153.1 (d, J = 97.5 Hz), 139.6 (d, J = 7.8 Hz), 139.1, 132.5 (d, J = 2.9 Hz), 132.4 (d, J = 2.9 Hz), 131.3 (d, J = 10.6 Hz), 131.1 (d, J = 10.3 Hz), 130.2 (d, J = 107.6 Hz), 129.6 (d, J = 108.5 Hz), 129.0, 128.7 (d, J = 4.9 Hz), 128.6 (d, J = 4.8 Hz), 127.3, 84.9 (d, J = 14.2 Hz), 21.1, 11.3 (d, J = 2.4 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 19.09. HRMS (EI) *m/z*: calcd for $C_{24}H_{22}O_3P$ $[M]^+$, 389.1301; found, 389.1320.

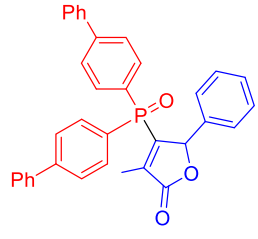
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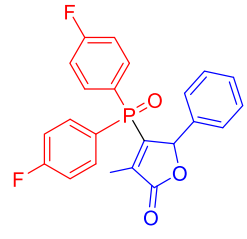
Scheme 3. Substrate Scope Relative to Diarylphosphine Oxide[ab]

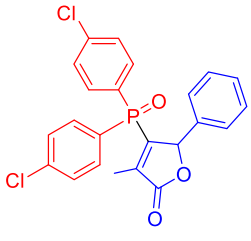


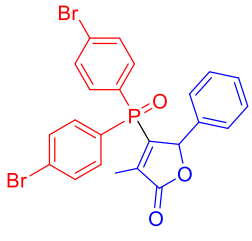

4a, 84%

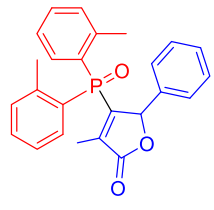

4b, 88%

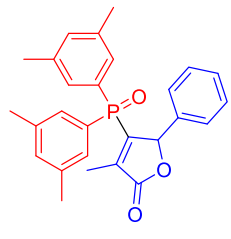

4c, 58%

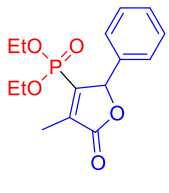

4d, 69%

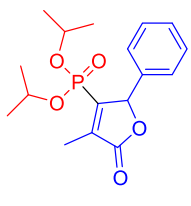

4e, 61%

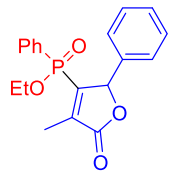

4f, 63%


4g, 80%


4h, 43%

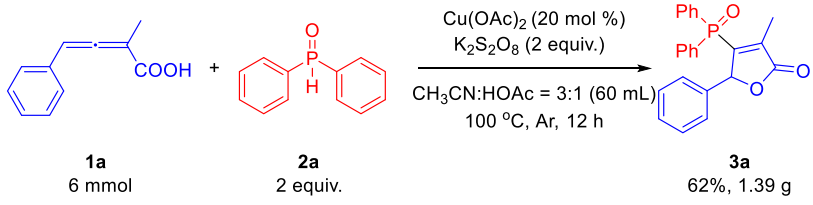

4i, 31%

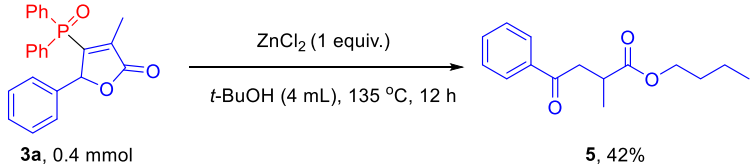

4j, 33%


4k, 41%

[a]Reaction conditions: **1a** (0.2 mmol), **2** (0.4 mmol), $K_2S_2O_8$ (2 equiv), $Cu(OAc)_2$ (20 mol %), and CH_3CN: HOAc = 3:1 (2.0 mL) at 100 °C for 12 h under Ar. [b]Isolated yield.

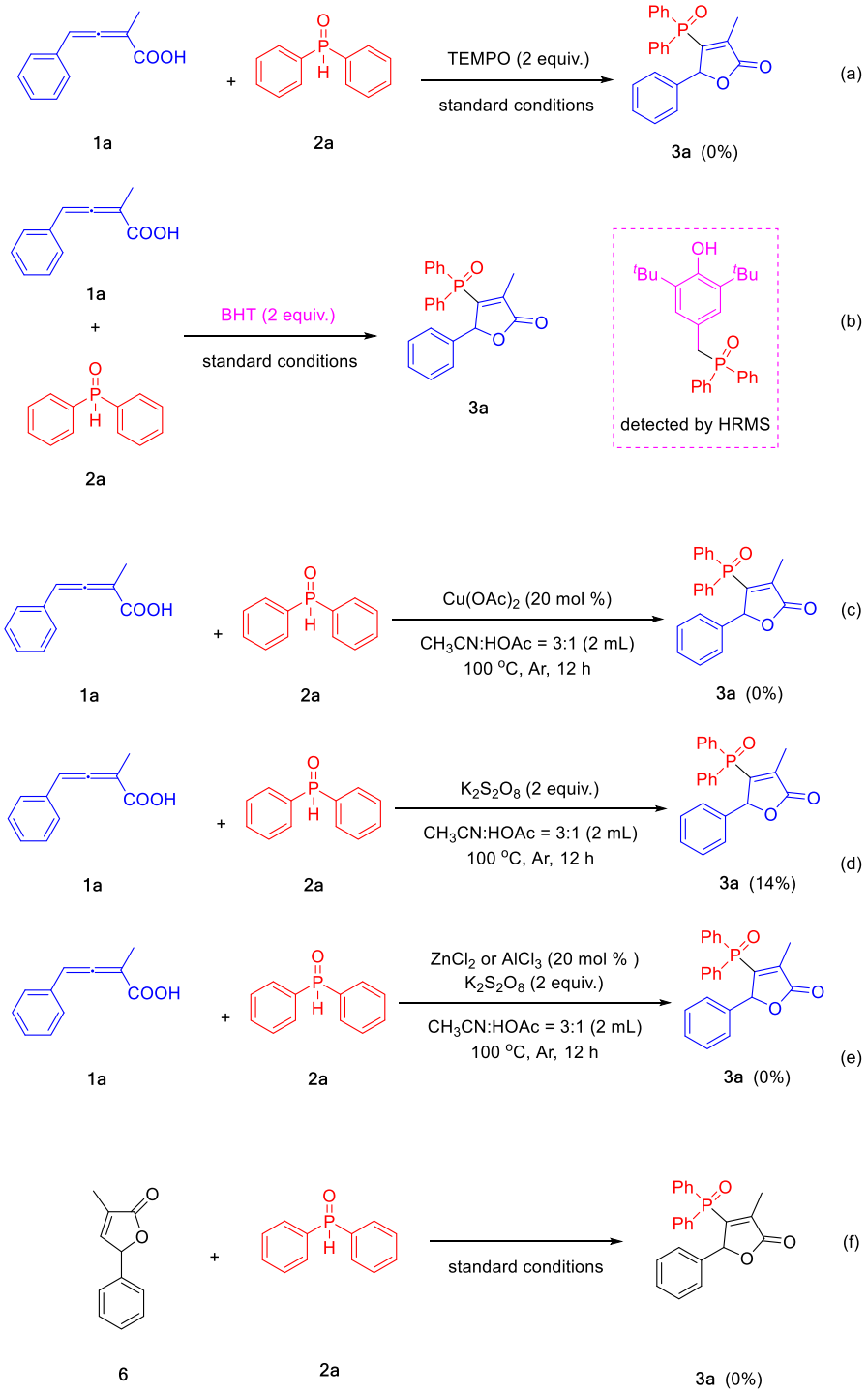
Scheme 4. Gram-Scale Experiment and Derivatization Reactions





https://doi.org/10.1021/acs.joc.1c00998
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Scheme 5. Control Experiments

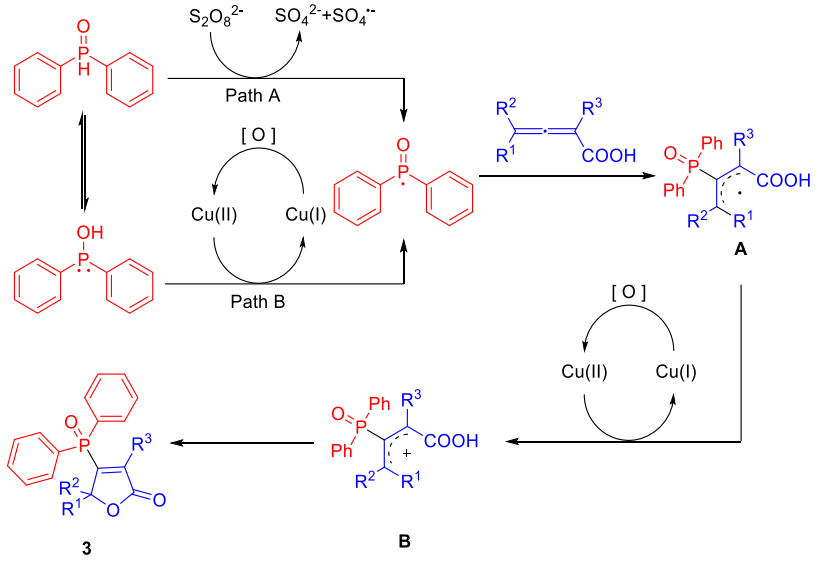


*5-(4-(tert-butyl)phenyl)-4-(diphenylphosphoryl)-3-methylfuran-2(5H)-one (**3c**).* The product **3c** was obtained as a white solid (36.1 mg, 42% yield, m.p. 210−211 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.52−7.34 (m, 10H), 7.08 (d, J = 8.4 Hz, 2H), 6.81 (d, J = 8.4 Hz, 2H), 6.03 (s, 1H), 1.85 (t, J = 2.2 Hz, 3H), 1.23 (s, 9H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 173.0 (d, J = 20.7 Hz), 153.1 (d, J = 97.9 Hz), 152.1, 139.2, 132.6 (d, J = 2.9 Hz), 132.4 (d, J = 2.9 Hz), 131.3 (d, J = 10.7 Hz), 131.0 (d, J = 10.2 Hz), 130.3 (d, J = 108.5 Hz), 129.6, 129.3 (d, J = 108.8 Hz), 128.7 (d, J = 8.6 Hz), 128.6 (d, J = 8.5 Hz), 127.0, 125.3, 84.8 (d, J = 14.1 Hz), 34.5, 31.1, 11.3 (d, J = 2.3 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 19.47. HRMS (EI) m/z: calcd for $C_{27}H_{28}O_3P$ $[M]^+$, 431.1771; found, 431.1780.

*4-(diphenylphosphoryl)-5-(4-methoxyphenyl)-3-methylfuran-2(5H)-one (**3d**).* The product **3d** was obtained as a white solid (56.6 mg, 70% yield, m.p. 150−151 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.58−7.33 (m, 10H), 6.78 (d, J = 8.6 Hz, 2H), 6.60 (d, J = 8.6 Hz, 2H), 5.96 (s, 1H), 3.73 (s, 3H), 1.87 (t, J = 2.2 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 172.8 (d, J = 20.7 Hz), 160.1, 153.0 (d, J = 97.5 Hz), 139.6 (d, J = 7.8 Hz), 132.6 (d, J = 2.8 Hz), 132.5 (d, J = 2.7 Hz), 131.3 (d, J = 10.6 Hz), 131.1 (d, J = 10.2 Hz), 130.3 (d, J = 108.4 Hz), 129.7 (d, J = 108.4 Hz), 128.8, 128.8 (d, J = 1.4 Hz), 128.6 (d, J = 1.4 Hz), 124.6, 113.8, 84.7 (d, J = 14.1 Hz), 55.2, 11.3 (d, J = 2.3 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 18.98. HRMS (EI) m/z: calcd for $C_{24}H_{22}O_4P$ $[M]^+$, 405.1250; found, 405.1252.

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Scheme 6. Proposed Reaction Mechanism



*4-(diphenylphosphoryl)-5-(4-fluorophenyl)-3-methylfuran-2(5H)-one (**3e**).* The product **3e** was obtained as a white solid (37.6 mg, 48% yield, m.p. 157−158 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.58−7.37 (m, 10H), 6.93−6.86 (m, 2H), 6.77 (t, J = 8.6 Hz, 2H), 6.06 (s, 1H), 1.82 (t, J = 2.2 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 172.6 (d, J = 20.5 Hz), 162.9 (d, J = 248.9 Hz), 153.1 (d, J = 97.6 Hz), 139.4 (d, J = 7.7 Hz), 132.8 (d, J = 2.8 Hz), 132.6 (d, J = 2.8 Hz), 131.3 (d, J = 10.7 Hz), 131.0 (d, J = 10.2 Hz), 130.3 (d, J = 108.5 Hz), 129.4 (d, J = 8.7 Hz), 129.3 (d, J = 108.7 Hz), 128.9 (d, J = 9.9 Hz), 128.8 (d, J = 3.2 Hz), 128.8 (d, J = 9.9 Hz), 115.4 (d, J = 21.9 Hz), 84.1 (d, J = 13.9 Hz), 11.3 (d, J = 2.2 Hz); ^{19}F NMR (376 MHz, $CDCl_3$) δ − 111.84; ^{31}P NMR (162 MHz, $CDCl_3$) δ 19.10. HRMS (EI) m/z: calcd for $C_{23}H_{19}FO_3P$ $[M]^+$, 393.1050; found, 393.1063.

*5-(4-chlorophenyl)-4-(diphenylphosphoryl)-3-methylfuran-2(5H)-one (**3f**).* The product **3f** was obtained as a white solid (66.9 mg, 82% yield, m.p. 167−168 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.56 (t, J = 6.6 Hz, 2H), 7.52−7.38 (m, 8H), 7.04 (d, J = 8.4 Hz, 2H), 6.85 (d, J = 8.4 Hz, 2H), 6.05 (s, 1H), 1.80 (t, J = 2.2 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 172.6 (d, J = 20.6 Hz), 153.2 (d, J = 97.6 Hz), 139.2 (d, J = 7.9 Hz), 135.1, 132.9 (d, J = 2.9 Hz), 132.6 (d, J = 2.8 Hz), 131.6, 131.4 (d, J = 10.7 Hz), 131.1 (d, J = 10.2 Hz), 130.4 (d, J = 108.6 Hz), 129.2 (d, J = 108.4 Hz), 128.9 (d, J = 12.8 Hz), 128.8 (d, J = 12.7 Hz), 128.8, 128.6, 84.1 (d, J = 13.6 Hz), 11.3 (d, J = 2.1 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 19.13. HRMS (EI) m/z: calcd for $C_{23}H_{19}ClO_3P$ $[M]^+$, 409.0755; found, 409.0757.

*5-(4-bromophenyl)-4-(diphenylphosphoryl)-3-methylfuran-2(5H)-one (**3g**).* The product **3g** was obtained as a white solid (70.5 mg, 78% yield, m.p. 186−187 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.60−7.36 (m, 10H), 7.18 (d, J = 8.4 Hz, 2H), 6.79 (d, J = 8.4 Hz, 2H), 6.04 (s, 1H), 1.79 (t, J = 2.2 Hz, 3H); ^{13}C{^{1}H}NMR (100 MHz, $CDCl_3$) δ 172.6 (d, J = 20.5 Hz), 153.0 (d, J = 97.3 Hz), 139.1 (d, J = 7.6 Hz), 132.8 (d, J = 2.5 Hz), 132.5 (d, J = 2.5 Hz), 132.0, 131.4, 131.3 (d, J = 10.6 Hz), 131.0 (d, J = 10.2 Hz), 130.3 (d, J = 108.6 Hz), 129.1 (d, J = 108.5 Hz), 129.0, 128.9 (d, J = 12.8 Hz), 128.7 (d, J = 12.9 Hz), 123.3, 84.0 (d, J = 13.6 Hz), 11.2 (d, J = 1.8 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 19.15. HRMS (EI) m/z: calcd for $C_{23}H_{19}BrO_3P$ $[M]^+$, 453.0250; found, 453.0242.

*4-(diphenylphosphoryl)-3-methyl-5-(4-(trifluoromethyl)phenyl)-furan-2(5H)-one (**3h**).* The product **3h** was obtained as a white solid (54.8 mg, 62% yield, m.p. 185−186 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.60−7.34 (m, 10H), 7.31 (d, J = 8.2 Hz, 2H), 7.09 (d, J = 8.2 Hz, 2H), 6.20 (s, 1H), 1.76 (t, J = 2.2 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 172.6 (d, J = 20.5 Hz), 153.2 (d, J = 97.2 Hz), 138.8 (d, J = 7.7 Hz), 137.3, 133.0 (d, J = 2.7 Hz), 132.7 (d, J = 2.7 Hz), 131.4 (d, J = 11.0 Hz), 131.0 (d, J = 10.2 Hz), 129.4 (d, J = 108.0 Hz), 129.0 (d, J = 12.7 Hz), 128.9 (d, J = 108.5 Hz), 128.8 (d, J = 12.6 Hz), 127.8, 125.3 (q, J = 3.8 Hz), 123.6 (q, J = 272.3 Hz), 83.9 (d, J = 13.4 Hz), 11.3 (d, J = 1.9 Hz); ^{19}F NMR (376 MHz, $CDCl_3$) δ − 62.96; ^{31}P NMR (162 MHz, $CDCl_3$) δ 19.38. HRMS (EI) m/z: calcd for $C_{24}H_{19}F_3O_3P$ $[M]^+$, 443.1018; found, 443.1030.

*5-(3-chlorophenyl)-4-(diphenylphosphoryl)-3-methylfuran-2(5H)-one (**3i**).* The product **3i** was obtained as a white solid (62.8 mg, 77% yield, m.p. 177−178 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.60−7.38 (m, 10H), 7.14−7.02 (m, 2H), 6.92 (d, J = 7.6 Hz, 1H), 6.71 (t, J = 1.9 Hz, 1H), 6.01 (s, 1H), 1.83 (t, J = 2.2 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 172.6 (d, J = 20.5 Hz), 153.0 (d, J = 96.7 Hz), 139.5 (d, J = 7.7 Hz), 135.1, 134.3, 132.9 (d, J = 2.7 Hz), 132.8 (d, J = 2.8 Hz), 131.4 (d, J = 10.6 Hz), 131.0 (d, J = 10.2 Hz), 130.3 (d, J = 109.3 Hz), 129.7, 129.3, 129.2 (d, J = 108.4 Hz), 128.9 (d, J = 12.8 Hz), 128.8 (d, J = 12.7 Hz), 127.6, 125.6, 84.0 (d, J = 13.7 Hz), 11.3 (d, J = 2.1 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 19.10. HRMS (EI) m/z: calcd for $C_{23}H_{19}ClO_3P$ $[M]^+$, 409.0755; found, 409.0770.

*4-(diphenylphosphoryl)-3-methyl-5-(o-tolyl)furan-2(5H)-one (**3j**).* The product **3j** was obtained as a white solid (28.7 mg, 37% yield, m.p. 161−162 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.65−7.55 (m, 3H), 7.50−7.38 (m, 5H), 7.35−7.30 (m, 2H), 7.10 (t, J = 7.4 Hz, 1H), 7.02−6.92 (m, 2H), 6.87 (d, J = 7.8 Hz, 1H), 6.23 (s, 1H), 2.07 (s, 3H), 1.93 (s, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 172.9 (d, J = 20.6 Hz), 152.8 (d, J = 97.2 Hz), 140.5 (d, J = 7.6 Hz), 137.4, 132.6 (d, J = 2.8 Hz), 132.5 (d, J = 2.9 Hz), 131.2 (d, J = 8.0 Hz), 131.1 (d, J = 7.7 Hz), 130.9, 130.7, 130.3 (d, J = 108.2 Hz), 129.5 (d, J = 108.3 Hz), 129.3, 128.8 (d, J = 12.6 Hz), 128.6 (d, J = 12.7 Hz), 127.5, 126.1, 81.8 (d, J = 14.1 Hz), 18.8, 11.5 (d, J = 2.5 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 18.43. HRMS (EI) m/z: calcd for $C_{24}H_{22}O_3P$ $[M]^+$, 389.1301; found, 389.1314.

*4-(diphenylphosphoryl)-5-(2-methoxyphenyl)-3-methylfuran-2(5H)-one (**3k**).* The product **3k** was obtained as a white solid (56.6 mg, 70% yield, m.p. 155−156 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.64−7.51 (m, 3H), 7.48−7.39 (m, 5H), 7.33−7.28 (m, 2H), 7.14 (t, J = 7.0 Hz, 1H), 6.89 (d, J = 7.4 Hz, 1H), 6.74 (t, J = 7.4 Hz, 1H), 6.51 (d, J = 11.4 Hz, 1H), 6.28 (s, 1H), 3.51 (d, J = 5.8 Hz, 3H), 1.96 (d, J = 2.4 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 173.1 (d, J = 20.8 Hz), 157.5, 151.7 (d, J = 97.9 Hz), 141.1 (d, J = 7.8 Hz), 132.4 (d, J = 2.8 Hz), 132.2 (d, J = 2.8 Hz), 131.2 (d, J = 7.4 Hz), 131.1 (d, J = 7.0 Hz), 130.8, 130.0 (d, J = 107.9 Hz), 129.5, 129.2 (d, J = 109.2 Hz), 128.6 (d, J = 12.6 Hz), 128.4 (d, J = 12.7 Hz), 120.6, 120.4, 110.6, 80.7 (d, J = 7.9 Hz), 55.0, 11.4 (d, J =

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2.5 Hz); ^{31}P NMR (162 MHz, Chloroform-*d*) δ 18.62. HRMS (EI) *m/z*: calcd for $C_{24}H_{22}O_4P$ $[M]^+$, 405.1250; found, 405.1256.

*5-(2-chlorophenyl)-4-(diphenylphosphoryl)-3-methylfuran-2(5H)-one (**3l**).* The product **3l** was obtained as a white solid (53.9 mg, 66% yield, m.p. 172−173 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.76−7.53 (m, 3H), 7.53−7.39 (m, 5H), 7.36−7.29 (m, 2H), 7.17−7.02 (m, 4H), 6.42 (s, 1H), 1.98 (t, *J* = 2.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 172.6 (d, *J* = 20.5 Hz), 153.0 (d, *J* = 96.7 Hz), 139.5 (d, *J* = 7.7 Hz), 135.1, 134.3, 132.9 (d, *J* = 2.7 Hz), 132.8 (d, *J* = 2.8 Hz), 131.4 (d, *J* = 10.6 Hz), 131.0 (d, *J* = 10.2 Hz), 130.3((d, *J* = 109.3 Hz), 129.7, 129.3, 129.2 (d, *J* = 108.4 Hz), 128.9 (d, *J* = 12.8 Hz), 128.8 (d, *J* = 12.7 Hz), 127.6, 125.6, 84.0 (d, *J* = 13.7 Hz), 11.3 (d, *J* = 2.1 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 17.92. HRMS (EI) *m/z*: calcd for $C_{23}H_{19}ClO_3P$ $[M]^+$, 409.0755; found, 409.0763.

*5-(2,5-dimethylphenyl)-4-(diphenylphosphoryl)-3-methylfuran-2(5H)-one (**3m**).* The product **3m** was obtained as a white solid (61.9 mg, 77% yield, m.p. 179−180 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.62−7.39 (m, 8H), 7.36 (t, *J* = 6.4 Hz, 2H), 6.87 (q, *J* = 8.6, 7.8 Hz, 2H), 6.53 (s, 1H), 6.23 (s, 1H), 2.08 (d, *J* = 3.4 Hz, 6H), 1.94 (t, *J* = 2.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 173.0 (d, *J* = 20.6 Hz), 153.0 (d, *J* = 99.5 Hz), 140.1 (d, *J* = 7.4 Hz), 135.4, 134.4, 132.6 (d, *J* = 2.1 Hz), 131.2 (d, *J* = 9.7 Hz), 131.1 (d, *J* = 9.3 Hz), 130.6, 130.4 (d, *J* = 107.4 Hz), 130.3, 130.1, 128.7 (d, *J* = 1.7 Hz), 128.6 (d, *J* = 1.7 Hz), 128.0, 81.9 (d, *J* = 13.8 Hz), 20.7, 18.4, 11.6; ^{31}P NMR (162 MHz, $CDCl_3$) δ 18.65. HRMS (EI) *m/z*: calcd for $C_{25}H_{24}O_3P$ $[M]^+$, 403.1458; found, 403.1476.

*5-(3,5-bis(trifluoromethyl)phenyl)-4-(diphenylphosphoryl)-3-methylfuran-2(5H)-one (**3n**).* The product **3n** was obtained as a white solid (53.0 mg, 52% yield, m.p. 182−183 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.58 (d, *J* = 7.4 Hz, 2H), 7.58−7.43 (m, 7H), 7.44−7.31 (m, 4H), 6.33 (s, 1H), 1.73 (t, *J* = 2.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 172.2 (d, *J* = 20.4 Hz), 152.9 (d, *J* = 97.1 Hz), 138.8 (d, *J* = 7.6 Hz), 136.5, 133.3 (d, *J* = 2.7 Hz), 132.9 (d, *J* = 2.5 Hz), 131.7 (q, *J* = 33.8 Hz), 131.3 (d, *J* = 10.9 Hz), 130.8 (d, *J* = 10.1 Hz), 130.3 (d, *J* = 109.1 Hz), 129.2 (d, *J* = 12.8 Hz), 128.8 (d, *J* = 12.5 Hz), 128.3 (d, *J* = 110.9 Hz), 127.6, 122.9 (dt, *J* = 7.6, 3.6 Hz), 122.6 (q, *J* = 273.0 Hz) , 83.0 (d, *J* = 13.0 Hz), 11.2; ^{19}F NMR (376 MHz, $CDCl_3$) δ −62.95; ^{31}P NMR (162 MHz, $CDCl_3$) δ 19.45. HRMS (EI) *m/z*: calcd for $C_{25}H_{18}F_6O_3P$ $[M]^+$, 511.0892; found, 511.0898.

*4-(diphenylphosphoryl)-3-methyl-5-(naphthalen-2-yl)furan-2(5H)-one (**3o**).* The product **3o** was obtained as a white solid (43.3 mg, 51% yield, m.p. 195−196 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.71 (d, *J* = 7.2 Hz, 1H), 7.57 (t, 3H), 7.47−7.35 (m, 9H), 7.29 (s, 1H), 7.25−7.15 (m, 3H), 6.98 (d, *J* = 8.4 Hz, 1H), 6.18 (s, 1H), 1.92 (t, *J* = 2.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 172.8 (d, *J* = 20.6 Hz), 153.1 (d, *J* = 97.1 Hz), 139.8 (d, *J* = 7.8 Hz), 133.4, 132.7 (d, *J* = 3.0 Hz), 132.6, 132.4 (d, *J* = 2.9 Hz), 131.3 (d, *J* = 10.7 Hz), 131.0 (d, *J* = 10.3 Hz), 130.6 (d, *J* = 9.7 Hz), 130.1 (d, *J* = 108.6 Hz), 129.9129.3 (d, *J* = 108.7 Hz), 128.8, 128.6 (d, *J* = 4.6 Hz), 128.4 (d, *J* = 5.6 Hz), 128.0 (d, *J* = 5.4 Hz), 127.5, 126.7, 126.2, 123.3, 85.0 (d, *J* = 14.1 Hz), 11.3 (d, *J* = 2.5 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 19.15. HRMS (EI) *m/z*: calcd for $C_{27}H_{22}O_3P$ $[M]^+$, 425.1301; found, 425.1314.

*4-(diphenylphosphoryl)-5-phenyl-3-propylfuran-2(5H)-one (**3p**).* The product **3p** was obtained as a white solid (57.9 mg, 72% yield, m.p. 221−222 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.56−7.29 (m, 10H), 7.16 (d, *J* = 7.4 Hz, 1H), 7.08 (t, *J* = 7.4 Hz, 2H), 6.84 (d, *J* = 7.2 Hz, 2H), 5.99 (s, 1H), 2.26 (t, *J* = 8.0 Hz, 2H), 1.57−1.44 (m, 1H), 1.43−1.29 (m, 1H), 0.75 (t, *J* = 7.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 172.3 (d, *J* = 20.7 Hz), 152.8 (d, *J* = 97.1 Hz), 144.1 (d, *J* = 8.2 Hz), 133.0, 132.5 (d, *J* = 2.5 Hz), 132.5 (d, *J* = 2.5 Hz), 131.3 (d, *J* = 10.5 Hz), 131.1 (d, *J* = 10.2 Hz), 130.3 (d, *J* = 108.1 Hz), 130.0 (d, *J* = 108.2 Hz), 129.1, 128.7 (d, *J* = 4.8 Hz), 128.5 (d, *J* = 4.9 Hz), 128.4, 127.3, 84.7 (d, *J* = 14.3 Hz), 27.8 (d, *J* = 2.2 Hz), 21.2 (d, *J* = 1.7 Hz), 14.1; ^{31}P NMR (162 MHz, $CDCl_3$) δ 18.63. HRMS (EI) *m/z*: calcd for $C_{25}H_{24}O_3P$ $[M]^+$, 403.1458; found, 403.1476.

*4-(diphenylphosphoryl)-3,5-dimethyl-5-phenylfuran-2(5H)-one (**3q**).* The product **3q** was obtained as a white solid (21.7 mg, 28% yield, m.p. 146−147 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.57 (t, *J* = 7.4 Hz, 1H), 7.54−7.42 (m, 5H), 7.34−7.28 (m, 4H), 7.24−7.20 (m, 2H), 7.14−7.10 (m, 3H), 2.20 (s, 3H), 1.61 (d, *J* = 2.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 172.5 (d, *J* = 21.1 Hz), 157.4 (d, *J* = 94.9 Hz), 136.9, 136.8 (d, *J* = 9.1 Hz), 132.7 (d, *J* = 2.6 Hz), 132.3 (d, *J* = 2.6 Hz), 131.7(d, *J* = 108.1 Hz), 131.4 (d, *J* = 10.8 Hz), 131.2 (d, *J* = 10.2 Hz), 129.4 (d, *J* = 108.1 Hz), 129.0 (d, *J* = 12.8 Hz), 128.5 (d, *J* = 12.7 Hz), 128.4, 128.2, 126.0, 90.5 (d, *J* = 12.8 Hz), 24.4, 11.6 (d, *J* = 1.9 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 20.99. HRMS (EI) *m/z*: calcd for $C_{24}H_{22}O_3P$ $[M]^+$, 389.1301; found, 389.1305.

*4-(diphenylphosphoryl)-3-methyl-5,5-diphenylfuran-2(5H)-one (**3r**).* The product **3r** was obtained as a white solid (29.7 mg, 33% yield, m.p. 187−188 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.52−7.43 (m, 2H), 7.41−7.31 (m, 12H), 7.24 (t, *J* = 7.6 Hz, 6H), 1.70 (d, *J* = 2.4 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 171.9 (d, *J* = 21.0 Hz), 156.9, 139.2 (d, *J* = 9.1 Hz), 137.8, 132.3 (d, *J* = 2.9 Hz), 131.5 (d, *J* = 10.5 Hz), 130.7 (d, *J* = 108.4 Hz), 128.7, 128.6, 128.5 (d, *J* = 12.8 Hz), 128.0, 95.5 (d, *J* = 11.3 Hz), 12.1 (d, *J* = 2.3 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 21.21. HRMS (EI) *m/z*: calcd for $C_{29}H_{24}O_3P$ $[M]^+$, 451.1458; found, 451.1477.

*4-(di-p-tolylphosphoryl)-3-methyl-5-phenylfuran-2(5H)-one (**4a**).* The product **4a** was obtained as a white solid (67.6 mg, 84% yield, m.p. 149−150 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.41−7.24 (m, 4H), 7.23−7.12 (m, 5H), 7.09 (t, *J* = 7.4 Hz, 2H), 6.86 (d, *J* = 7.0 Hz, 2H), 5.99 (s, 1H), 2.39 (d, *J* = 9.2 Hz, 6H), 1.89 (t, *J* = 2.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 173.0 (d, *J* = 20.4 Hz), 153.6 (d, *J* = 97.0 Hz), 143.3 (d, *J* = 2.9 Hz), 143.1 (d, *J* = 2.8 Hz), 139.3 (d, *J* = 7.7 Hz), 133.0, 131.3 (d, *J* = 10.9 Hz), 131.1 (d, *J* = 10.6 Hz), 129.5 (d, *J* = 4.2 Hz), 129.4 (d, *J* = 4.1 Hz), 129.0, 128.3, 127.4, 127.0 (d, *J* = 111.1 Hz), 126.4 (d, *J* = 111.1 Hz), 85.0 (d, *J* = 14.0 Hz), 21.6 (d, *J* = 1.5 Hz), 21.6 (d, *J* = 1.4 Hz), 11.3 (d, *J* = 2.4 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 19.61. HRMS (EI) *m/z*: calcd for $C_{25}H_{24}O_3P$ $[M]^+$, 403.1458; found, 403.1470.

*4-(bis(4-methoxyphenyl)phosphoryl)-3-methyl-5-phenylfuran-2(5H)-one (**4b**).* The product **4b** was obtained as a white solid (76.4 mg, 88% yield, m.p. 131−132 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.41−7.27 (m, 4H), 7.19 (t, *J* = 7.4 Hz, 1H), 7.11 (t, *J* = 7.4 Hz, 2H), 6.92−6.80 (m, 6H), 6.00 (s, 1H), 3.82 (d, *J* = 8.0 Hz, 6H), 1.90 ((t, *J* = 2.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 172.9 (d, *J* = 20.3 Hz), 162.8 (d, *J* = 2.8 Hz), 162.7 (d, *J* = 3.0 Hz), 153.9 (d, *J* = 97.8 Hz), 139.0 (d, *J* = 7.7 Hz), 133.1 (d, *J* = 12.1 Hz), 133.1, 132.9 (d, *J* = 11.7 Hz), 129.0, 128.2, 127.4, 121.5 (d, *J* = 115.8 Hz), 120.7 (d, *J* = 115.5 Hz), 114.3 (d, *J* = 1.5 Hz), 114.1 (d, *J* = 1.4 Hz), 84.9 (d, *J* = 14.0 Hz), 55.3, 55.2, 11.2 (d, *J* = 2.2 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 19.28. HRMS (EI) *m/z*: calcd for $C_{25}H_{24}O_5P$ $[M]^+$, 435.1356; found, 435.1370.

*4-(di([1,1′-biphenyl]-4-yl)phosphoryl)-3-methyl-5-phenylfuran-2(5H)-one (**4c**).* The product **4c** was obtained as a white solid (61.0 mg, 58% yield, m.p. 191−192 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.65−7.54 (m, 10H), 7.53−7.38 (m, 8H), 7.18 (t, *J* = 7.4 Hz, 1H), 7.09 (t, *J* = 7.6 Hz, 2H), 6.91 (d, *J* = 7.6 Hz, 2H), 6.06 (s, 1H), 1.98 (t, *J* = 2.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 172.8 (d, *J* = 20.7 Hz), 152.9 (d, *J* = 98.2 Hz), 145.5 (d, *J* = 2.9 Hz), 145.4 (d, *J* = 2.9 Hz), 140.0 (d, *J* = 7.8 Hz), 139.4 (d, *J* = 6.7 Hz), 133.0, 131.9 (d, *J* = 11.0 Hz), 131.7 (d, *J* = 10.7 Hz), 129.1, 129.0 (d, *J* = 2.7 Hz), 128.4, 127.5, 127.5(d, *J* = 4.1 Hz), 127.2, 85.0 (d, *J* = 14.2 Hz), 11.5 (d, *J* = 2.4 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 19.52. HRMS (EI) *m/z*: calcd for $C_{35}H_{28}O_3P$ $[M]^+$, 527.1771; found, 527.1789.

*4-(bis(4-fluorophenyl)phosphoryl)-3-methyl-5-phenylfuran-2(5H)-one (**4d**).* The product **4d** was obtained as a white solid (56.6 mg, 69% yield, m.p. 147−148 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.54−7.44 (m, 2H), 7.43−7.36 (m, 2H), 7.22 (t, *J* = 7.2 Hz, 1H), 7.16−7.02 (m, 6H), 6.89 (d, *J* = 7.0 Hz, 2H), 5.99 (s, 1H), 1.90 (t, *J* = 2.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$) δ 172.5 (d, *J* = 20.9 Hz), 166.5 (t, *J* = 3.4 Hz), 164.0 (t, *J* = 3.4 Hz), 152.4 (d, *J* = 99.4 Hz), 140.0 (d, *J* = 8.1 Hz), 133.8 (dd, *J* = 12.1, 8.8

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Hz), 133.6 (dd, J = 11.7, 9.2 Hz), 132.7, 129.4, 128.5, 127.3, 126.0 (dd, J = 112.2, 2.9 Hz), 125.3 (dd, J = 112.2, 3.3 Hz), 116.4 (d, J = 13.9 Hz), 116.2 (d, J = 13.6 Hz), 84.8(d, J = 14.3 Hz), 11.4 (d, J = 2.2 Hz); ^{19}F NMR (376 MHz, Chloroform-d) δ − 104.40 (d, J = 52.6 Hz). ^{31}P NMR (162 MHz, $CDCl_3$) δ 17.67. HRMS (EI) m/z: calcd for $C_{23}H_{18}F_2O_3P$ $[M]^+$, 411.0956; found, 411.0958.

*4-(bis(4-chlorophenyl)phosphoryl)-3-methyl-5-phenylfuran-2(5H)-one (**4e**).* The product **4e** was obtained as a white solid (53.9 mg, 61% yield, m.p. 159−160 °C, hexane/EA = 2:1); ^{1}H NMR (600 MHz, $CDCl_3$) δ 7.43−7.37 (m, 4H), 7.32 (d, J = 16.2 Hz, 4H), 7.22 (t, J = 11.4 Hz, 1H), 7.13 (t, J = 11.4 Hz, 2H), 6.87 (d, J = 10.6 Hz, 2H), 5.96 (s, 1H), 1.92 (d, J = 2.8 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 172.4 (d, J = 21.1 Hz), 152.0 (d, J = 99.6 Hz), 140.4 (d, J = 7.9 Hz), 139.6 (d, J = 3.4 Hz), 139.5 (d, J = 3.4 Hz), 132.6, 132.5 (d, J = 11.6 Hz), 132.3 (d, J = 11.2 Hz), 129.4, 129.3 (d, J = 1.5 Hz), 129.2 (d, J = 1.5 Hz), 128.6, 128.2 (d, J = 110.6 Hz), 127.6 (d, J = 110.5 Hz), 127.38, 84.8 (d, J = 14.4 Hz), 11.5 (d, J = 2.5 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 17.79. HRMS (EI) m/z: calcd for $C_{23}H_{18}Cl_2O_3P$ $[M]^+$, 443.0365; found, 443.0361.

*4-(bis(4-bromophenyl)phosphoryl)-3-methyl-5-phenylfuran-2(5H)-one (**4f**).* The product **4f** was obtained as a white solid (66.8 mg, 63% yield, m.p. 179−180 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.55 (dd, J = 8.4, 2.4 Hz, 2H), 7.50 (dd, J = 8.4, 2.4 Hz, 2H), 7.33 (d, J = 8.4 Hz, 1H), 7.30 (d, J = 8.4 Hz, 1H), 7.28−7.19 (m, 3H), 7.14 (t, J = 7.6 Hz, 2H), 6.87 (d, J = 9.6 Hz, 2H), 5.94 (s, 1H), 1.92 (t, J = 2.2 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 172.3 (d, J = 21.1 Hz), 151.9 (d, J = 99.5 Hz), 140.5 (d, J = 7.8 Hz), 132.7, 132.6 (d, J = 11.4 Hz), 132.4 (d, J = 11.1 Hz), 132.3 (d, J = 1.8 Hz), 132.2 (d, J = 1.9 Hz), 129.5, 128.8 (d, J = 109.9 Hz), 128.6, 128.3 (d, J = 3.6 Hz), 128.3 (d, J = 109.8 Hz), 128.2 (d, J = 3.6 Hz), 127.4, 84.8 (d, J = 14.4 Hz), 11.5 (d, J = 2.6 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 17.94. HRMS (EI) m/z: calcd for $C_{23}H_{18}Br_2O_3P$ $[M]^+$, 530.9355; found, 530.9344.

*4-(di-o-tolylphosphoryl)-3-methyl-5-phenylfuran-2(5H)-one (**4g**).* The product **4g** was obtained as a white solid (64.3 mg, 80% yield, m.p. 137−138 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.42−7.31 (m, 2H), 7.20 (d, J = 7.6 Hz, 2H), 7.12−6.99 (m, 8H), 6.90−6.81 (m, 1H), 6.19 (s, 1H), 2.52 (s, 3H), 2.41 (s, 3H), 1.75 (t, J = 2.2 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 173.0 (d, J = 20.2 Hz), 153.5 (d, J = 95.7 Hz), 143.5 (d, J = 8.2 Hz), 142.9 (d, J = 8.7 Hz), 138.4 (d, J = 7.3 Hz), 133.1, 132.6 (d, J = 2.6 Hz), 132.3 (d, J = 2.8 Hz), 132.2 (d, J = 10.7 Hz), 132.0 (d, J = 11.0 Hz), 131.8 (d, J = 3.1 Hz), 131.7, 128.8, 128.8 (d, J = 105.5 Hz), 128.1, 127.2 (d, J = 107.8 Hz), 126.8, 125.7 (d, J = 13.7 Hz), 125.4 (d, J = 13.3 Hz), 84.9 (d, J = 13.5 Hz), 21.4 (d, J = 4.4 Hz), 21.3 (d, J = 4.5 Hz), 10.8 (d, J = 2.2 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 26.50. HRMS (EI) m/z: calcd for $C_{25}H_{24}O_3P$ $[M]^+$, 403.1458; found, 403.1470.

*4-(bis(3,5-dimethylphenyl)phosphoryl)-3-methyl-5-phenylfuran-2(5H)-one (**4h**).* The product **4h** was obtained as a white solid (37.0 mg, 43% yield, m.p. 181−182 °C, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.25−7.10 (m, 5H), 7.05 (s, 1H), 7.02 (s, 1H), 6.99 (s, 1H), 6.96 (s, 1H), 6.83 (d, J = 7.3 Hz, 2H), 5.95 (s, 1H), 2.26 (d, J = 19.9 Hz, 12H), 1.93 (t, J = 2.2 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 173.1 (d, J = 20.4 Hz), 153.6 (d, J = 95.7 Hz), 139.7 (d, J = 7.4 Hz), 138.55, 138.42, 134.3 (d, J = 3.0 Hz), 134.3 (d, J = 3.0 Hz), 132.9, 129.7 (d, J = 107.7 Hz), 129.4(d, J = 107.6 Hz) , 129.0, 128.9 (d, J = 10.7 Hz), 128.7 (d, J = 10.3 Hz), 128.2, 127.5, 85.0 (d, J = 14.0 Hz), 21.25, 21.18, 11.4 (d, J = 2.4 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 20.10. HRMS (EI) m/z: calcd for $C_{27}H_{28}O_3P$ $[M]^+$, 431.1771; found, 431.1776.

*Diethyl(4-methyl-5-oxo-2-phenyl-2,5-dihydrofuran-3-yl)-phosphonate (**4i**).* The product **4i** was obtained as a colorless liquid (19.2 mg, 31% yield, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.39−7.35 (m, 3H), 7.29−7.24 (m, 2H), 5.96−5.94 (m, 1H), 3.99−3.82 (m, 3H), 3.76−3.66 (m, 1H), 2.26 (dd, J = 3.2, 2.0 Hz, 3H), 1.21 (t, J = 7.2 Hz, 3H), 1.07 (t, J = 7.2 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 172.8 (d, J = 29.0 Hz), 148.2 (d, J = 195.1 Hz), 139.8 (d, J = 11.8 Hz), 133.9, 129.5, 128.6, 127.4, 84.2 (d, J = 18.0 Hz), 62.5 (d, J = 6.4 Hz), 62.4 (d, J = 6.3 Hz), 16.2 (d, J = 6.3 Hz), 15.9 (d, J = 6.7 Hz), 11.0 (d, J = 2.4 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 7.71. HRMS (EI) m/z: calcd for $C_{15}H_{20}O_5P$ $[M]^+$, 311.1043; found, 311.1058.

*Diisopropyl(4-methyl-5-oxo-2-phenyl-2,5-dihydrofuran-3-yl)-phosphonate (**4j**).* The product **4j** was obtained as a colorless liquid (23.8 mg, 33% yield, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.39−7.32 (m, 3H), 7.25 (d, J = 2.6 Hz, 2H), 5.95 (s, 1H), 4.62−4.38 (m, 2H), 2.28−2.23 (m, 3H), 1.24 (d, J = 6.2 Hz, 3H), 1.19−1.11 (m, 6H), 1.01 (d, J = 6.2 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 173.1 (d, J = 28.9 Hz), 149.5 (d, J = 196.6 Hz), 139.0 (d, J = 12.0 Hz), 134.1, 129.3, 128.5, 127.5, 84.4(d, J = 17.8 Hz), 71.8 (d, J = 26.8 Hz), 71.7 (d, J = 26.8 Hz), 23.9 (d, J = 3.8 Hz), 23.8, 23.8, 23.4 (d, J = 4.8 Hz), 11.0 (d, J = 2.2 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 5.39. HRMS (EI) m/z: calcd for $C_{17}H_{23}NaO_5P$ $[M]^+$, 361.1175; found, 361.1177.

*Ethyl(4-methyl-5-oxo-2-phenyl-2,5-dihydrofuran-3-yl)(phenyl)-phosphinate (**4k**).* The product **4k** was obtained as a colorless liquid (28.1 mg, 41% yield, hexane/EA = 2:1); ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.53−7.28 (m, 5H), 7.23 (t, J = 7.2 Hz, 2H), 7.15−7.05 (m, 3H), 6.00 (d, J = 6.0 Hz, 1H), 4.11−3.82 (m, 2H), 2.21 (d, J = 10.8 Hz, 3H), 1.26 (dt, J = 37.2, 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 172.8 (dd, J = 24.3, 15.0 Hz), 151.5 (dd, J = 137.8, 28.7 Hz), 139.1 (dd, J = 159.6, 11.9 Hz), 133.4 (d, J = 16.2 Hz), 132.8 (t, J = 2.3 Hz), 131.1 (dd, J = 35.8, 10.9 Hz), 129.3 (d, J = 10.9 Hz), 129.1 (dd, J = 141.5, 60.2 Hz), 128.5 (dd, J = 13.7, 4.2 Hz), 128.4 (d, J = 6.4 Hz), 127.7 (d, J = 15.5 Hz), 84.2 (dd, J = 26.1, 15.4 Hz), 61.4 (t, J = 5.3 Hz), 16.1 (dd, J = 13.9, 6.9 Hz), 10.8 (dd, J = 23.4, 2.2 Hz); ^{31}P NMR (162 MHz, $CDCl_3$) δ 23.27 (d, J = 72.4 Hz). HRMS (EI) m/z: calcd for $C_{19}H_{20}O_4P$ $[M]^+$, 343.1094; found, 343.1104.

*Open Ring of 4-(diphenylphosphoryl)-3-methyl-5-phenylfuran-2(5H)-one (**3a**).* To an oven-dried 10 mL pressure tube were added 4-(diphenylphosphoryl)-3-methyl-5-phenylfuran-2(5H)-one (**3a**) (0.4 mmol), $ZnCl_2$(0.4 mmol), and t-BuOK (0.4 mL). After stirring at 135 °C (temperature of the oil bath) overnight, the mixture was concentrated under reduced pressure. The residue was then purified by flash chromatography (PE/EA = 4:1) to give **5** (42 mg, 42% yield) as a colorless liquid; ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.04−7.88 (m, 2H), 7.57 (t, J = 7.4 Hz, 1H), 7.46 (t, J = 7.6 Hz, 2H), 4.10 (td, J = 6.8, 2.4 Hz, 2H), 3.49 (dd, J = 17.6, 7.8 Hz, 1H), 3.13 (td, J = 7.4, 5.6 Hz, 1H), 3.01 (dd, J = 17.6, 5.6 Hz, 1H), 1.78−1.52 (m, 2H), 1.38 (dt, J = 15.0, 7.4 Hz, 2H), 1.28 (d, J = 7.2 Hz, 3H), 0.92 (t, J = 7.4 Hz, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 198.1, 176.0, 136.7, 133.1, 128.6, 128.0, 64.5, 41.9, 35.1, 30.6, 19.1, 17.3, 13.7. HRMS (EI) m/z: calcd for $C_{15}H_{20}NaO_3$ $[M]^+$, 271.1305; found, 271.1317.

General Procedure for the Synthesis of Substrates 6.[20] Under N_2, a solution of **1a** (380 mg, 2.0 mmol) and CuCl (0.08 mmol, 4 mol %) in methanol (20 mL) was stirred under reflux for 2 h. After evaporation, the residues were purified via flash chromatography on silica gel with PE/EA (10:1) as the eluent to provide the 3-methyl-5-phenyl-2(5H)-furanone (226 mg, 65% yield) as a colorless liquid; ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.42−7.32 (m, 3H), 7.28−7.21 (m, 2H), 7.16−7.10 (m, 1H), 5.86 (s, 1H), 1.99 (s, 3H); ^{13}C{^{1}H} NMR (100 MHz, $CDCl_3$) δ 174.2, 148.4, 135.0, 129.2, 128.9, 128.8, 126.3, 82.0, 10.5.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.1c00998.

Experimental details and spectral data for all products (PDF).

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Notes

The authors declare no competing financial interest.

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